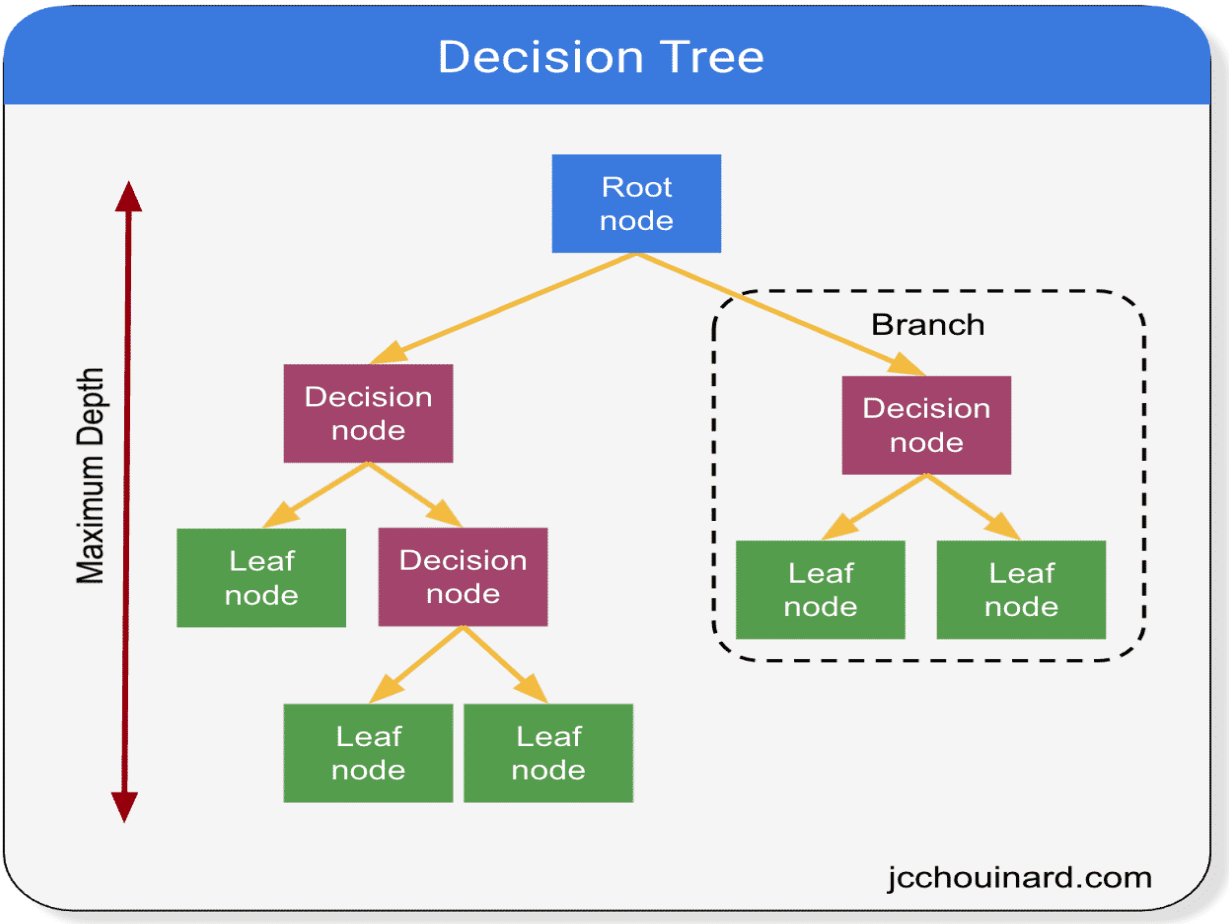
Decision Trees

1. Introduction to Decision Trees

What are Decision Trees?

A **Decision Tree** is a supervised machine-learning algorithm used for classification and regression tasks. It works by breaking down a dataset into smaller subsets based on feature values, forming a tree-like structure. Each internal node represents a decision based on a feature, branches represent possible outcomes, and leaf nodes represent final predictions.

Decision Trees are widely used due to their **interpretability, ease of use, and ability to handle both numerical and categorical data**.



2. Binary Decision Trees

What is a Binary Decision Tree?

A **Binary Decision Tree** is a decision tree where each node has at most two children. It is used in classification tasks where each decision splits the dataset into two groups.

Binary Decisions

Binary decisions refer to **splitting the data into two parts based on a given condition** (e.g., feature > threshold). The tree structure grows by recursively applying this splitting process until a stopping criterion is met.

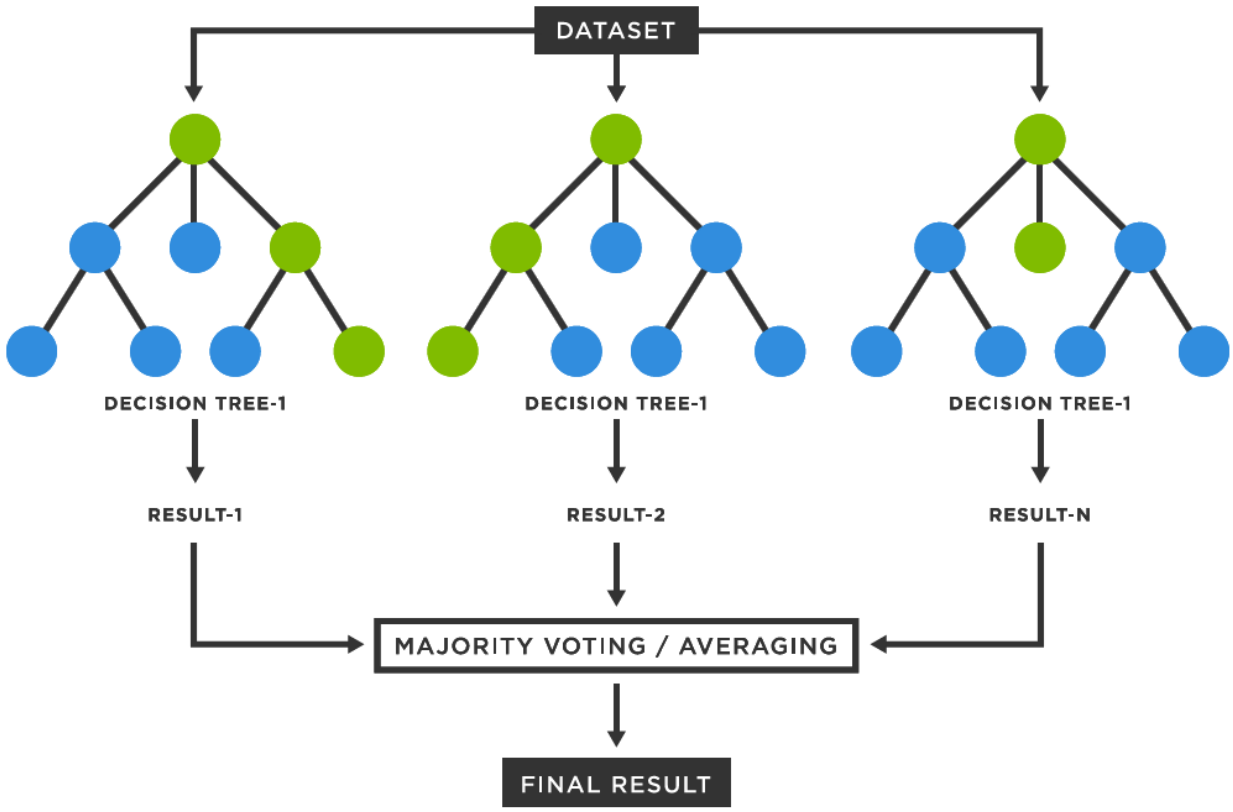
3. CART Algorithm (Classification and Regression Trees)

What is the CART Algorithm?

The **CART Algorithm** is the foundation for constructing decision trees. It is a **greedy algorithm** that recursively selects the best feature and threshold to split the dataset.

Steps in the CART Algorithm:

1. **Select the Best Feature** - Identify the feature that results in the best split.
2. **Split the Dataset** - Divide the dataset into two subsets based on the selected feature.
3. **Repeat Recursively** - Apply the same process to each subset until a stopping condition is met.
4. **Assign Class Labels** - Assign a class label to each leaf node based on the majority class.

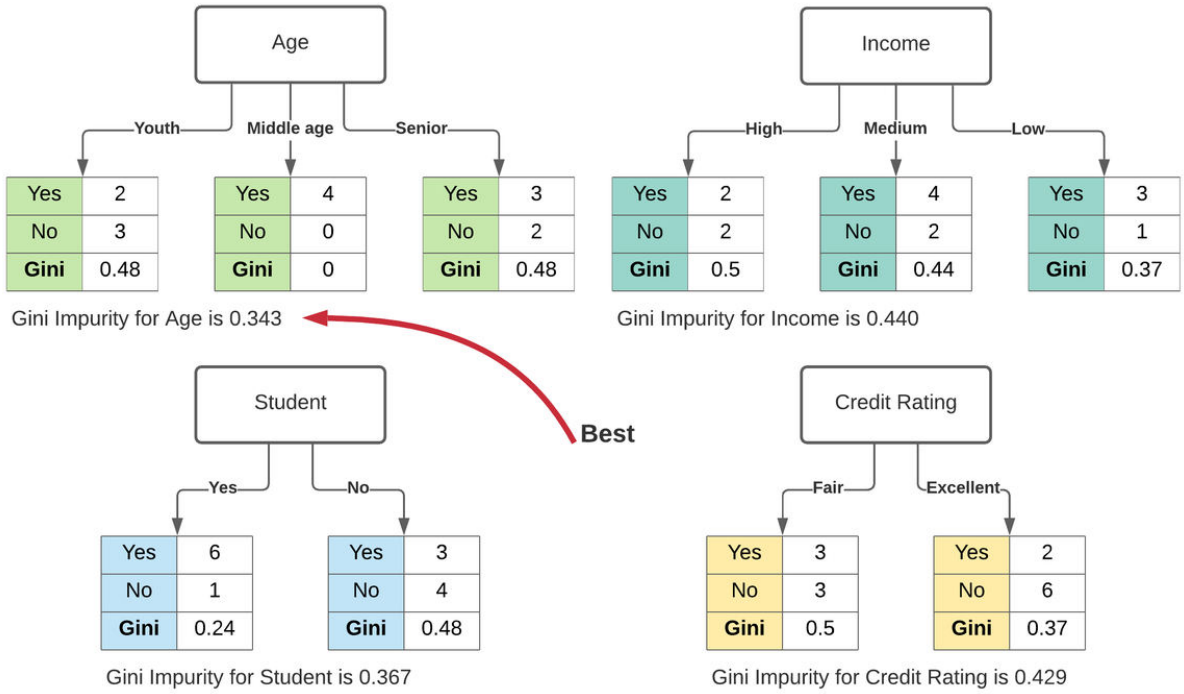


4. Impurity Measures

Impurity measures determine how well a feature split separates the classes. Lower impurity means a better split.

1. Gini Impurity Index

- Measures the **probability of misclassification** if a random sample is assigned a label.
- Formula: $Gini = 1 - \sum p_i^2$ Where p_i is the probability of class i.
- A Gini index of **0** represents a pure split, meaning all elements belong to one class.



2. Cross-Entropy (Information Gain/Entropy)

- Measures the **uncertainty of a system before and after splitting** the data.
- Formula: $Entropy = -\sum p_i \log_2 p_i$
- The **higher the entropy, the greater the disorder**, and the lower the entropy, the better the split.

3. Misclassification Impurity Index

- Measures the **fraction of incorrect classifications** at a node.
- Formula: $Misclassification = 1 - \max(p_i)$ where p_i is the proportion of data points belonging to class i.

5. Feature Importance

What is Feature Importance?

Feature importance determines how relevant each feature is in making predictions. It helps to understand which features contribute the most to the decision-making process.

How is Feature Importance Measured?

1. **Gini Importance**: Reduction in Gini impurity when a feature is used for splitting.

2. **Information Gain**: Reduction in entropy after a split.
3. **Permutation Importance**: Change in model performance when a feature's values are shuffled randomly.

6. Decision Tree Classification with Scikit-learn

Implementation Example:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Sample Dataset
X = np.array([[2, 3], [10, 15], [5, 7], [6, 8], [9, 11]])  # Features
y = np.array([0, 1, 0, 0, 1])  # Labels

# Splitting data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Decision Tree Classifier
dt = DecisionTreeClassifier(criterion='gini', max_depth=3, random_state=42)
dt.fit(X_train, y_train)

# Predictions
y_pred = dt.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred))
```

Explanation:

1. **Data Preparation**: A sample dataset is created.
2. **Splitting Data**: The dataset is split into training and testing sets.
3. **Model Training**: A decision tree classifier is trained on the data.
4. **Predictions & Accuracy**: The trained model predicts the class labels for test data, and accuracy is calculated.

7. Advantages & Disadvantages

Advantages of Decision Trees

Easy to Understand - Simple and interpretable model.

Handles Both Numerical & Categorical Data - Works well with different types of data.

Requires Little Data Preprocessing - No need for feature scaling.

Works with Large Datasets - Scales well with large amounts of data.

Disadvantages of Decision Trees

Prone to Overfitting - Can create overly complex trees without pruning.

Sensitive to Noisy Data - Small changes in data can lead to a different tree structure.

Biased Splitting - If class distribution is imbalanced, it can create biased trees.

8. Conclusion

Decision Trees are a powerful and interpretable classification technique. Understanding binary trees, the CART algorithm, impurity measures, and feature importance helps in building efficient and reliable models. Implementing decision trees in **scikit-learn** is straightforward, making it a valuable tool for machine learning practitioners.

7. Kaggle and GitHub Projects

Kaggle Projects:

1. https://www.kaggle.com/code/dmilla/introduction-to-decision-trees-titanic-dataset
2. https://www.kaggle.com/code/tasnimniger/loan-default-prediction

GitHub Projects:

1. Decision Tree Implementation from Scratch
2. Scikit-learn Decision Tree Example

---

8. Interview Questions & Answers

What is a Decision Tree?

A supervised learning algorithm used for classification and regression.

How does a Decision Tree split data?

It splits based on impurity measures like Gini Index and Entropy.

What are the advantages of Decision Trees?

Easy to interpret, handles both numerical and categorical data.

What is the difference between Gini Impurity and Entropy?

Gini measures misclassification probability, Entropy measures information gain.

How do you prevent overfitting in Decision Trees?

Use pruning, set a maximum depth, or use ensemble methods.

What is pruning in Decision Trees?

Reducing the size of a tree by removing less important branches.

What is the difference between a Decision Tree and a Random Forest?

A Decision Tree is a single tree; a Random Forest is an ensemble of multiple trees.

How do Decision Trees handle missing values?

By assigning the most frequent value or using surrogate splits.

What is the role of Feature Importance in Decision Trees?

It helps determine which features contribute the most to classification.

Can Decision Trees be used for regression tasks?

Yes, Decision Trees can be used for both classification and regression.